



NAUVAL ZULFIKAR

Applied ML / NLP Engineer · Responsible AI candidate Bandung, Indonesia (relocatable to Aarhus, 2026-08-01) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264

[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

Visa status: UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

MSc Business Analytics graduate from Aston University with a First Class degree, specialising in machine learning and NLP applications. Extensive experience in developing AI systems for decision-making and optimisation in various sectors, including banking, government, and start-ups. MSc dissertation focused on blockchain-integrated LLM analysis for supply chain systems, supervised by Dr. Ammar Al-Bazi. Proven track record in enhancing operational efficiency through data-driven solutions.

EDUCATION

Aston University, Birmingham, UK

MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan

BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- **Distinctions:** Business Data Analysis, Consumer Behaviour, Marketing Research

INFORMAL EDUCATION

Indonesia AI

NLP Engineer Training · EQF Level 5 · May – Nov 2023

Contributed to the *2024 Indonesia Election Aspect-Based Sentiment Analyser* project.

Distinctions: Python Programming, Text Summarisation, Voice Recognition.

Qarir Generator (Dubai, UAE)

Data Science Global Accelerator · EQF Level 5 · Apr – Jun 2023

Led sentiment-analysis development for the *Pfizer Public Sentiment Analysis* project.

Distinctions: Python, SQL, Pandas/NumPy.

RESEARCH-RELEVANT PROJECTS

Mini Digital Twin for Integrated Passenger-Freight Transport (May 2026)

Three-layer Digital Twin prototype with live folium map and interactive dashboards. Smoke-tested: 45% acceptance, 62% terminal utilisation.

github.com/nauvalZulfikar/Mini-Digital-Twin

LLM-Generated Adaptive Shipper Decision Rules (May 2026)

End-to-end RAG system using gpt-4o-mini for generating decision rules from traveller profiles.

github.com/nauvalZulfikar/LLM-Shipper-Profiles

VRPOD MILP Optimisation (May 2026)

MILP formulation for VRPOD problem, achieving 38.5% cost reduction on a toy instance.

github.com/nauvalZulfikar/VRP-MILP

Crowd-Shipping ABM Behavioural Sweep (May 2026)

450-run parameter sweep with significant findings on supply effect.

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung

Data Scientist & Project Manager · Sep 2022 – Present

Automated ETL pipelines and built self-service dashboards, achieving 45% faster project delivery and 20% data-efficiency gain.

Syncwell, Birmingham, UK

Marketing Data Scientist · Jul – Sep 2024

Implemented predictive ML and personalisation strategies, resulting in a 40% CTR increase and 22% follower growth.

PCOS Challenge, Birmingham, UK

Strategic Data Scientist Specialist · Jun – Aug 2024

Developed customer attribution models and interactive visualisations, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta

Lead Data Scientist · Nov 2019 – Aug 2022

Developed NLP models for credit analysis and risk scoring, achieving a 5.3% investment return.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Comparative evaluation of multiple ML regressors for retail sales forecasting accuracy.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API, RAG, sentence-transformers

Languages/Infrastructure: Python, SQL, R, PySpark, FastAPI, Streamlit

Mathematical/Formal: MILP, Agent-Based Modelling, Discrete-Event Simulation, Bayesian methods

Governance: Model validation, model reliability, audit-grade documentation, AWS, Docker, Git

LANGUAGES

Indonesian (*Native*) · **English** (*Fluent, IELTS 7.0 overall*) · **Japanese** (*Intermediate*)

REFEREES

Dr. Ammar Al-Bazi

Senior Lecturer, Aston Business School, Aston University

Email: a.al-bazi@aston.ac.uk

Dr. Viktor Pekar

Lecturer, Aston Business & Social Sciences

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